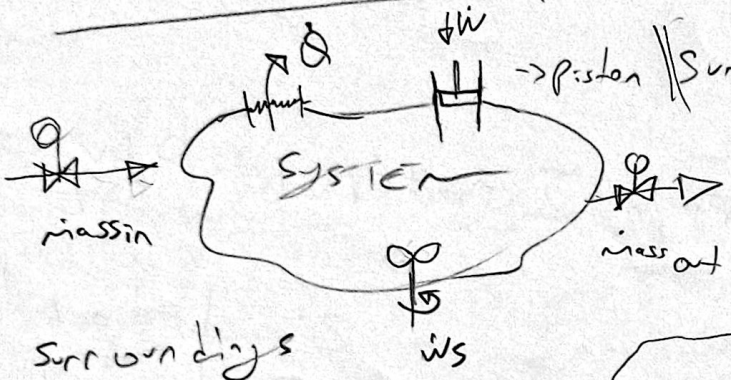
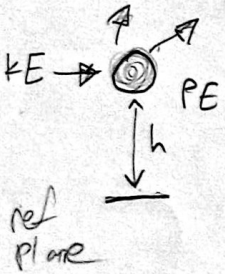
m1 T

Thermo - 5.60

Thermo - 1

Term





Energy

Energy in transit due is a ΔT

Energy in transit due to movement

KE, PE, U ... internal

Universe

system

surroundings

- Mass exchange
- heat exch [$\dot{Q}$]
- work exch.

→ diathermal wall

→ { - flow
- surface boundary movement
- shaft

Temperature

Pressure

EQUILIBRIUM

State

1 - Mass is conserved (conservation)

$t = t \qquad t = \Delta t + t$

$$\frac{dm}{dt} = \dot{m}_{in} - \dot{m}_{out} \qquad \boxed{\frac{dm}{dt} = \sum \dot{m}_{in} - \sum \dot{m}_{out}}$$

2 - The energy is conserved

$$\not{E} = U + PE + KE$$

$$\frac{dE}{dt} = \sum \dot{m}_{in} E_{in} - \sum \dot{m}_{out} \cdot E_{out} + \dot{Q} + \dot{W}$$

$$(?? \Rightarrow \dot{W} = \dot{W}_s + \sum \dot{m}_{in} (PV)_{in} - \sum \dot{m}_{out} (PV)_{out} - P\frac{dV}{dt}$$

video bitecek

2. sayfa

$$\delta w = F \cdot dx = P \cdot A \cdot \frac{dx}{dv}$$

$$\frac{dE}{dt} = \sum \dot{m}_{in} \hat{E}_{in} - \sum \dot{m} \hat{E}_{out} + \dot{Q} + \dot{W}_s + \sum \dot{m}_{in} (P\hat{v})_{in} - \sum \dot{m}_{out} (P\hat{v})_{out} - P \cdot \frac{dv}{dt}$$

Adiabatic, $\dot{Q} = 0$

$\dot{Q} = 0$ adiabatic

A → B

$t = 0$ $T_A > T_B$

$t = \infty$ $T_A = T_B$ Thermal Equilibrium.

open system ⇒ mi = dynamic closed sys. mi = 0

intensive properties ⇒ sistem büyüklüğüne bağlı değil.

Extensive properties → büyüklüğe bağlı.

MIT - THERMO

Temp

Sıfırıncı Law — T - common sense

1st Law — U - Yalıcan bi enerji

2nd Law — S — Entropi 0°K de %100 olur. → enerjinin kalitesini azaltığı yönündedir.

3rd Law — 0°K 'e ulaşamazsın.

eş-kesişim/iç içe geçmiş hali

Kontrol hacmi

mass, energy flow

No mass yes energy flow

open system

closed system

Isolated system

no mass, no energy

Definition

System ⟶ Surroundings

(Kontrol hacmi, K.Hacmi) Boundary (Sınır) (Kontrol yüzeyi)

Des. System (eg. labirent) → C

extensive → V, m büyüklüğe bağlı değil

intensive values → T, P

- Isıl denge
- Mekanik — Basınç farklı noktalarda eşit ve değişmez
- Kimyasal — eşit kimyasal bileşim

state veriables (durum değişkenleri)

Path
- Reversible (tersinir) → yavaş.
- Irreversible →

Ders

A ⟷ B 0. yasa

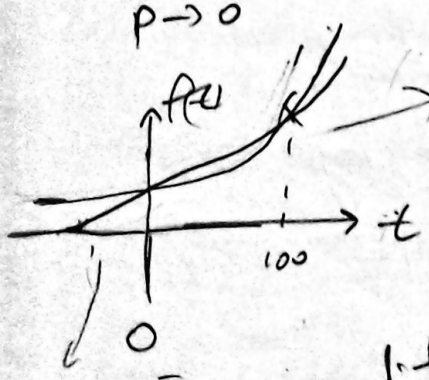
too → equilibrium

Lecture 2

Ideal Gaz Hanımefendi!

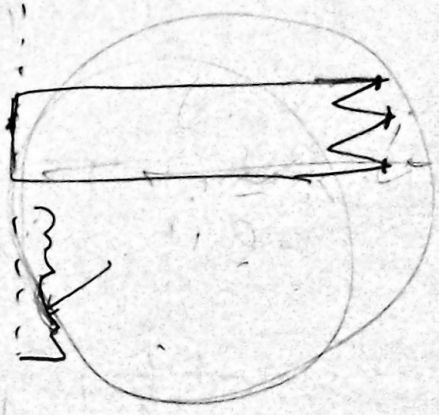
$\lim_{p \to 0} (p\bar{V}) = \text{constant} = f(t)$ (Sabit)

- Substance - gaz.
- Property = $\lim_{p \to 0} (p.\bar{V})$
- boiling point (100°C)
- freezing point (0°C)

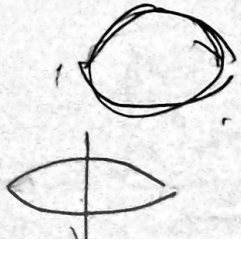


$f(t)=0 \Rightarrow p\bar{V}=0$

Negatif basınç hacim olmayacağından bir minimum değeri olduğu bilgisi.

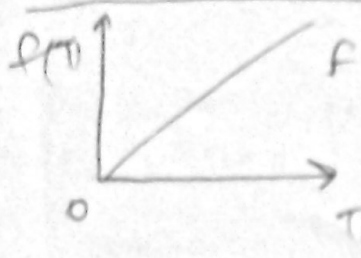






Lecture-2-Dem

IDEAL GAZ THEOREM



$RT = P\bar{V}$ ideal gz.

Real Gaz

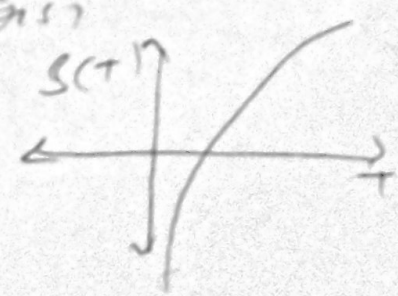
(a) Sikıştırılabilirlik $z = \frac{\bar{V}_{real}}{\bar{V}_{ideal}}$

$pV = R \cdot T \cdot z$

(b) virial genişleme

$$\frac{p\bar{V}_{real}}{RT} = z = 1 + \frac{B(T)}{\bar{V}_{real}} + \frac{C(T)}{\bar{V}^2_{real}}$$

→ 2. virial sayısı



1) van der vaals durum denklemi

$$\left(P + \frac{a}{\bar{V}^2}\right)(\bar{V} - b) = RT$$

$P \cdot \bar{V} = RT$ → gerçek gördüğü hacim değer atomların kapladığı alandan dolayı

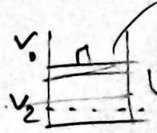
$P(\bar{V} - b) = RT$ düzeltme hacmi

$P = \frac{RT}{\bar{V} - b} - \frac{a}{\bar{V}^2}$ $a > 0$, çekim varsa

---

Work $= F \cdot l$ kuvvet × mesafe

Expansion work (sıkıştırma işi)



$V_1 < V_2$

$F = P \cdot A$ $\quad W = -F \cdot l$

$= -P \cdot A \cdot l$

$\Rightarrow P \cdot \Delta V$

$\Delta V < 0$

$W > 0$

pozitif işitir.

Lecture-2 - Devam-3

11.02.24

→ tam bir diff olmadığı anlamına geliyor.

Sınırları bilmemiz gerekiyor çünkü belirlenebilmek için.

$$\delta W = -P_{ext} \cdot dV$$

(P_{ext} (1,2))

$$\Rightarrow W = \int_1^2 \delta W = -\int_1^2 P_{ext} \cdot dV$$

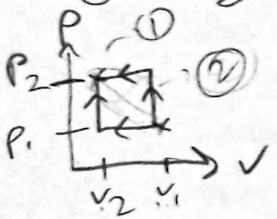
Argon $(g, P_1, V_1) = Ar(g, P_2, V_2)$

$V_1 > V_2$ $\quad P_1 < P_2$

① $V_1 \to V_2$, $P_1 \to P_2$

② $P_1 \to P_2$, $V_1 \to V_2$

(Tersinirler)



Tersinir olduğunda $\Rightarrow P_{ext} = P_{in}$

$$W_1 = -\int_{V_1}^{V_2} P_1 dV - \cancel{\int_{V_2}^{V_2} P \cdot dV}^{\to 0}$$

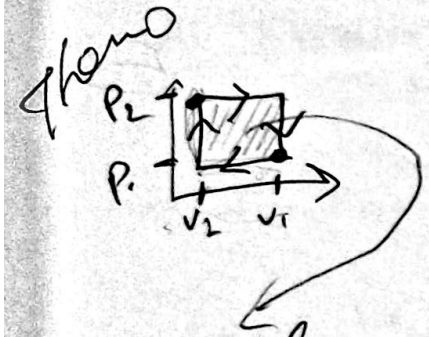
$$W_1 = -P_1 (V_2 - V_1) > 0$$

$$W_2 = -\cancel{\int_{V_1}^{V_1} P_1 dV}^{\to 0} - \int_{V_1}^{V_2} P_2 dV$$

$$W_2 = -P_2 (V_2 - V_1) = > 0$$

$$\boxed{W_2 > W_1}$$

Yapılan iş yola bağlıdır.



$W_{total} = W_1 - W_2$

$= P_1(V_1 - V_2) - P_2(V_1 - V_2)$

$= (P_1 - P_2)(V_1 - V_2) < 0$

W_{total}

* aynı noktaya dönüş sırasında alınan iş yapılan iş olursa. → kapalı yol

$W = \oint dW \neq 0$

Lecture-3

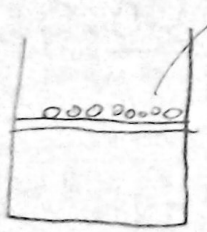
Thermo Heat Capacity

Thermo is

Isothermal Gas Expansions

4. do a reversibly

$P = P_{ext}$



$$W_{rev} = \int_{V_1}^{V_2} P\,dV$$

$$P = \frac{nRT}{V}$$

$$W_{rev} = \int_{V_1}^{V_2} \frac{nRT}{V}\,dV = -nRT \int_{V_1}^{V_2} \frac{dV}{V}$$

$$W = -$$

$$= -n \cdot R \cdot T \cdot \ln \frac{V_2}{V_1}$$

Lec-3-Devam

Thermo

max out. is by rev path

$\Delta U = q + w$

Const V (sabit)

$dw = 0 \quad du = dq$

$du = C_v dT + \left(\frac{\partial U}{\partial V}\right)_T dV$

Adyabatik

$dq = 0, \; du = dw$

her

rev

her işte değil

Joule's free expansions Deneyi

$\left(\frac{dT}{dV}\right)_U = 0$ $\quad du = C_v dT$

$\left(\frac{\partial U}{\partial V}\right)_T$ aslında sıfır değildir.

İDEAL GAZLAR için sıfırdır.

$\Delta U = 0$ All izotermal ideal gaz tepkimeleri için

$q = -w$

Thermo

Lecture - 4

Recall

$u(T) \Rightarrow du = C_v dT = dq_{sbt\ Hacim}$

$H(T,P)$ [Entalpi] $\quad H = u + pV$

$syst.(P_1, T_1, V_1) = sys(P_2, T_2, V_2)$

$\Delta u = q + w = q_{sp} - P\Delta V$

q_{sp} → sabit basınç denk

Sıcaklık değişimi sab. Hacim

$\Delta u + P\Delta V = q_{sp}$

$\Delta u + \Delta(PV) = q_{sp}$

$\Delta(u + PV) = q_{sp}$

$\Delta H = q_{sp}$ (constant) $\Rightarrow dH = dq_{sp}$

Constant Pressure → 0

$dH = \left(\frac{\partial H}{\partial T}\right)_P dT + \left(\frac{\partial H}{\partial P}\right)_T dP$

$\left(\frac{\partial H}{\partial P}\right)_T$ → sabit sıcaklık denk

Tersinir, sabit basınç

$dH = dq_{sp}$

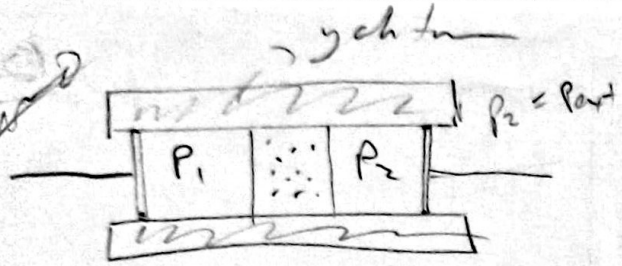
Per $\left(\frac{\partial H}{\partial T}\right)_P \Rightarrow dH = dq_{cp}$, $dH = \left(\frac{\partial H}{\partial T}\right)_P dT = dq_p = C_p dT$

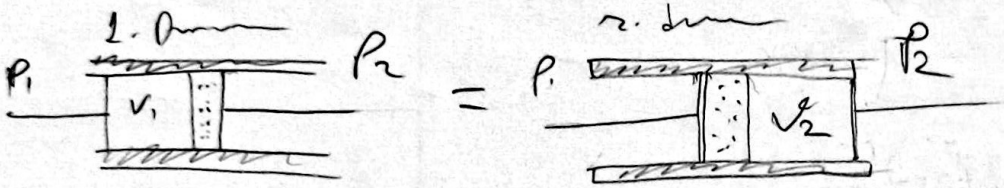
$\left(\frac{\partial H}{\partial T}\right)_P = C_p$

Joule - Thompson deneyi

$= \left(\frac{\partial H}{\partial P}\right)_T$ izenthalpic

Piston





$\Delta q = 0$
adyabatik

$H_1 = W_1 = P_1 V_1$ $\qquad$ $H_2 = W_2 = -P_2 V_2$

$W_{Total} = W_1 + W_2$

$W_T = P_1 V_1 - P_2 V_2$ $\qquad$ $= -\Delta(PV) = \Delta U$

$\Delta H = \Delta(u + PV)$

$= \Delta u + \Delta(PV)$

$= -\Delta(PV) + \Delta(PV) = 0 \neq \partial q_{cp}$

doğru ama zorunlu değil

Sabit Entalpi Genişlemesinin Tersinir değil Tersinmez bir süreç.

$dH \neq \partial q_{cpV}$

tersinir x

Lecture 4 Dersi

$\left(\frac{dH}{dT}\right)_P$, $\left(\frac{dH}{dP}\right)_T$?

$dH = C_P dT + \left(\frac{dH}{dP}\right)_T dP = 0$

$\left(\frac{dH}{dT}\right)_P = C_P$

→ Silebileceğim bir şey.

$\left(\frac{dH}{dP}\right)_T = -C_P \left(\frac{dT}{dP}\right)_H = \left(\frac{\Delta T}{\Delta P}\right)_H$

$\left(\frac{dH}{dP}\right)_T = -C_P \mu_{JT}$

ideal gaz

$U(T) = PV = nRT$

$H = U + PV = U(T) + nRT$

no P depend

$\left(\frac{dH}{dP}\right)_T = 0$ ⇒ $\mu_{JT} = 0$

Van der Vaals

$$\left(P + \frac{a}{\bar{V}^2}\right)(\bar{V} - b) = RT \Rightarrow \left(\frac{\partial H}{\partial P}\right)_T \propto b - \frac{a}{RT}$$

Proportional Constant

$$\left(\frac{\partial H}{\partial P}\right)_T \propto -\mu_{JT} \Rightarrow \boxed{\left(\frac{\partial T}{\partial P}\right)_H \propto \frac{a}{RT} - b}$$

Joule-Thomson değinde ideal gaz gibi davranış sergilediklerini söyleyebiliriz.

$\mu_{JT} < 0$

8.01

Termo MIT

$$\frac{a}{RT_{inv}} - b = 0 \Rightarrow T_{inv} = \frac{a}{Rb}$$

$T > T_{inv}$ için $\frac{a}{RT} - b < 0 \Rightarrow \left(\frac{dT}{dP}\right)_H < 0$

Basınç sabitlenirse sıcaklığı düşer.

$T < T_{inv}$ .... $\left(\frac{dT}{dP}\right)_H > 0$

Mutlak için $T_{inv} >> 300K$

İstisnalar
↳ Hidrojen = 193K
↳ Helyum = 53K

$C_p - C_v$ ilişkisi

$$\bar{C}_p = \bar{C}_v + R$$

~~$P \cdot V = nRT$~~

$P = \frac{nRT}{V}$

Hangi saltlarda MIT

lec 5

öğrenilenler

adiabatic → $dq = 0$

Rev → $dw = -p_{ex} dV = -p dV$

Ideal Gas → $dU = C_V dT$ $\quad pV = nRT$

I st Law → $dU = dq + dw = -p dV$ } eşitlersek → $C_V dT = -p dV$

$$C_V \ln\left(\frac{T_1}{T_2}\right) = -R \ln\left(\frac{V_1}{V_2}\right) \Leftarrow C_V \int_{T_1}^{T_2} \frac{dT}{T} = -R \int_{V_1}^{V_2} \frac{dV}{V}$$

$$\Rightarrow \ln\left(\frac{T_1}{T_2}\right)^{C_V} = \ln\left(\frac{V_2}{V_1}\right)^{R}$$

ikisi değişken gelsin diye "p" den kurtulmuştuk.

$$C_V \frac{dT}{T} = -R \frac{dV}{V}$$

V-arası

lec 5 Devam

Tlem 13
MIT

$\left(\frac{T_2}{T_1}\right) = \left(\frac{V_1}{V_2}\right)^{R/C_V}$

$\left(\frac{T_2}{T_1}\right) = \left(\frac{V}{V_2}\right)^{\gamma-1}$

$\frac{R}{C_V} = \frac{(C_P - C_V)}{C_V} = \frac{C_P}{C_V} - 1$

$\frac{C_P}{C_V} \equiv \gamma$

-1 yönlü bir sayının negatif bir sayıya dönüşmesi

$T = \frac{PV}{R} \Rightarrow \left(\frac{T_2}{T_1}\right) = \left(\frac{V_1}{V_2}\right)^{\gamma-1} \Rightarrow \left(\frac{P_2 V_2 / R}{P_1 V_1 / R}\right) = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$

$\Rightarrow \left(\frac{V_1}{V_2}\right)^{\gamma} = \frac{P_2}{P_1} \Rightarrow P_1 V_1^{\gamma} = P_2 V_2^{\gamma} = PV^{\gamma} = C$

Monotomic Ideal Gaz

$C_V = \frac{3}{2} R$

$C_P = \frac{5}{2} R$

$\gamma = \frac{5}{3}$

Argon monatomic ideal gazdır.

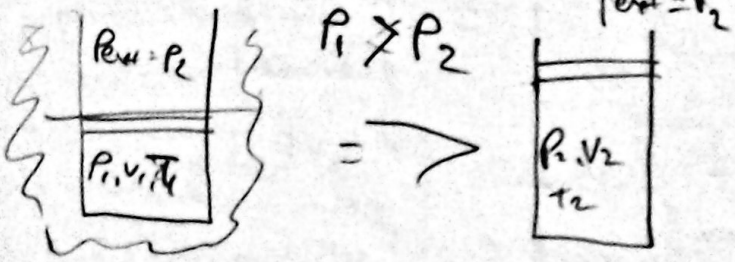
$\left(\frac{T_2}{T_1}\right) = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$ gaz

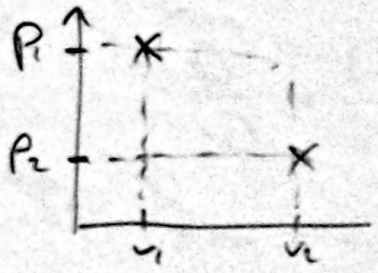
Argon için $\frac{5}{3} - 1 > 0$

$V_2 > V_1$ ise $T_1 > T_2$

isothermal V > normal c V

Irreversible - Adiabatic





$\delta q = 0$ since adiabatic

$\delta w = -P_2 dV$

$dU = C_v dT$ ideal gas

I^{st} / $dU = \delta w + \delta q \Rightarrow dU = -P_2 dV$

$T_2(C_v + R) = T_1(C_v + \frac{P_2}{P_1} R)$

$\frac{P_2}{P_1} < 1$ since $P_1 > P_2$

$T_2 < T_1$

$(-W_{irr}) < (-W_{rev})$

Lec - 6

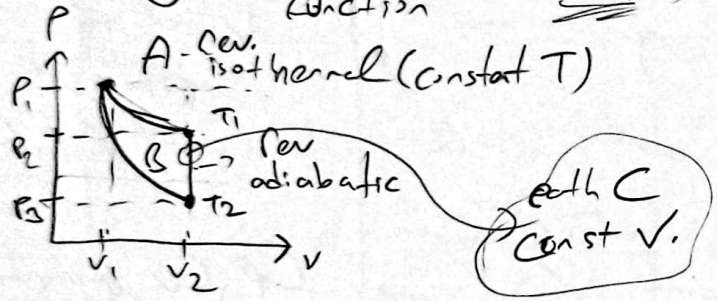
MET Therm: 0

Thermodynamic cycles

ΔU , q,
ΔH , w

ideal gas

"special function" $dS = \int \frac{dq}{T}$



Path A Const T.

$\Delta U_A = 0$ const T old. ve ideal gas old.dan

$\Delta H_A = 0$ [RTV] ideal old. $\Delta U = \partial q - w = 0 \Rightarrow \Delta U$

$\partial w^{rev} = -p dV = -RT_1 \frac{dV}{V}$

$w_A = -\int RT_1 \frac{dV}{V} = -R T_1 \ln \frac{V_2}{V_1}$

Cevreye yapılan iş, Cizilen cevreyi ihtiva eden

$q_A = -w_A = RT_1 \ln \frac{V_2}{V_1}$

$S_A = \frac{RT_1 \ln \frac{V_2}{V_1}}{T_1}$

Path B adiabatic

$q_B = 0$

$dU = C_V dT$ $\quad dU_B = C_V (T_2 - T_1)$ (w_B)

$dH = C_p dT$ $\quad dH_B = C_p (T_2 - T_1)$

$dS_B = \int \frac{dq_B}{T} = 0$

Path C Const V dedik $w_C = 0$ dedik

$\Delta U_C = C_V (T_1 - T_2) = q_C$

$\Delta H_C = C_p (T_1 - T_2)$

$dS = \int \frac{dq_C}{T} = \int_{T_2}^{T_1} \frac{dU}{T} = \int_{T_2}^{T_1} \frac{C_V dT}{T} = C_V \ln \frac{T_1}{T_2}$

A yada B ve C, ile gösterirsek

U ve H eşit oldu için bu örnekte "0"

q ve W eşit değil cünkü burda durum fonksiyonları değişir,

Special Fnch.

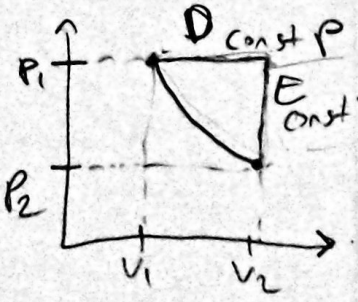
(A) $R \ln \frac{V_2}{V_1}$ (B+C) $= C_v \ln \frac{T_1}{T_2}$

Adiabatic Rev Path $\Rightarrow \frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{R/C_v}$

$\left(\frac{T_1}{T_2}\right)^{C_v} = \left(\frac{V_2}{V_1}\right)^{R}$

$\Delta S_A = \ln\left(\frac{V_2}{V_1}\right)^R$ $\Delta S_{B+C} = \ln\left(\frac{T_1}{T_2}\right)^{C_v}$

$S_A = S_{B+C}$ durum fonk. gibi davranışa yolden bağımsız



S_{D+E}

$S_D \neq S_E \neq 0$

Path D

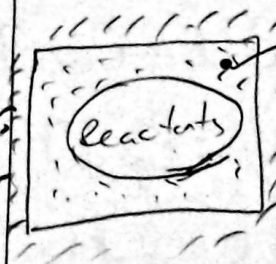
$\Delta U_D = C_v \Delta T = C_v (T_1 - T_2)$ (0)

$\Delta H_D = C_p (T_1 - T_2) =$

Thermochemistry

Rxn = reaksiyon

Calorimetry ?



Adiabatic const V

$\Delta H = 0$ Gizli $\Rightarrow$ const V.

$\Delta U = q$

(I) $\text{Reac}(T_1) + \text{Cal}(T_1) \xrightarrow{\text{adiabatic const V, rev}} \text{Prod}(T_2) + \text{Cal}(T_2)$

(II) isothermal const V. rev $\text{Reac}(T_1) = \text{Prod}(T_1) + \text{Cal}(T_1)$

(III) const V. rev. T değişir

2. Ödev

Entalpi ile ΔH ve ΔU hesaplama yolları

Kalorimetre ile ilgili sorular

AÇ formülleri.

C_p, C_v ayrıntılı araştır.

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(18)

(I) $\Delta U = q + w$

$\Delta U = 0$

(II) $\Delta U = q = \int_{T_1}^{T_2} C_V \, dT$

$\approx \int_{T_1}^{T_2} C_V^{cal} \, dT = C_V^{cal} \Delta T$

(III) $\Delta U = -\Delta U_2 = \Delta U_{reac}$

$\Delta U_{reac} = -C_V^{cal} \Delta T$

↓ bomb calorimeter ↓ adiabatic system

$H = U + pV$

$\Delta H = \Delta U + \Delta(pV)$

$\approx \Delta U + \Delta(nRT)$

$= \Delta U + RT_1 \Delta n_{gas}$ ← ideal gas

$\Delta H_{reac}(T_1) \approx -C_V^{cal} \Delta T + RT_1 \Delta n_{gas}$

Ex:

$2FeS_2(s) + \frac{11}{2} O_2(g) = Fe_2O_3(s) + 4SO_2(g)$

$T_1 = 298K$; 0 ; $-824 \frac{kJ}{mol}$; $-297 \frac{kJ}{mol}$

$\Delta H_f = -180 \frac{kJ}{mol}$

$\Delta H_{rx} = \approx 824 + 4(-297) - 2(-180)$

$= -1.652 kJ$

start with 1 mol FeS_2

$C_V = 10 kJ/K$ $\Delta U = -1648 kJ$

$\Delta H = -824 kJ$

$\Delta T = -824 kJ / 10 kJ/K$

$\Delta T = 82K$

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